Autocorrelation

How a vector overlaps its own shifted copies, and why HRR decoding depends on it

James Pustejovsky

Brandeis University
Department of Computer Science

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Why This Deck

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Bind $\mathbf{u}$ and $\mathbf{v}$, then decode with $\mathbf{u}$:

$$\mathbf{t} = \mathbf{u} \circledast \mathbf{v} = (4, 5, 2, 5), \qquad \mathbf{u} \# \mathbf{t} = (19, 13, 17, 15)$$

That is not $\mathbf{v} = (2, 0, 1, 1)$, only close to it: cosine 0.88. The whole gap is explained by one object, the **autocorrelation** of the role, $\mathbf{u} \# \mathbf{u}$: how $\mathbf{u}$ overlaps with shifted copies of itself.

Three questions

1. What is an autocorrelation, and how do you compute one by hand?
2. How does a role's autocorrelation decide what decoding returns?
3. Which roles decode exactly, and why do random roles at large n decode well enough?

Overlap at an Offset

Take a vector $\mathbf{x}$. Rotate a copy of it left by j places, and take the dot product of the copy with the original. That number is the overlap of $\mathbf{x}$ with itself at offset j:

$$(\mathbf{x} \# \mathbf{x})_j = \sum_k x_k \, x_{(k+j) \bmod n}$$

The list of all n overlaps, offsets 0 to $n-1$, is the **autocorrelation** of $\mathbf{x}$: the correlation of $\mathbf{x}$ with itself.

Reminder: why $+j$ means "rotate left"

Slot k of the copy reads entry $k+j$ of $\mathbf{x}$, the entry j places ahead, so every entry moves j places toward the front. For $\mathbf{x} = (x_0, x_1, x_2, x_3)$ and $j = 1$, the copy is (x_1, x_2, x_3, x_0).

Each entry asks one question: if I slide $\mathbf{x}$ along itself by j, how much does it still line up?

Computing $\mathbf{u} \# \mathbf{u}$ by Hand

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

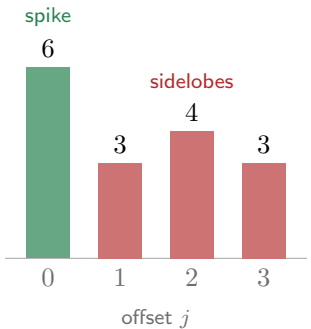
Rotate $\mathbf{u} = (1, 2, 0, 1)$ left by j, and dot with $\mathbf{u}$:

offset j	**$\mathbf{u}$ rotated left by j, dotted with $\mathbf{u}$**	$(\mathbf{u} \# \mathbf{u})_j$	
0	$(1,2,0,1) \cdot (1,2,0,1) = 1+4+0+1$	6	spike
1	$(2,0,1,1) \cdot (1,2,0,1) = 2+0+0+1$	3	sidelobe
2	$(0,1,1,2) \cdot (1,2,0,1) = 0+2+0+2$	4	sidelobe
3	$(1,1,2,0) \cdot (1,2,0,1) = 1+2+0+0$	3	sidelobe

So $\mathbf{u} \# \mathbf{u} = (6, 3, 4, 3)$. Offset 0 compares $\mathbf{u}$ with itself, unshifted: the **spike**, always $\|\mathbf{u}\|^2$. Every other offset compares $\mathbf{u}$ with a shifted copy: an accidental overlap, called a **sidelobe**.

The names come from radar and antennas: a strong main lobe, with smaller side lobes around it.

The Shape of an Autocorrelation



- The spike measures how much of $\mathbf{u}$ there is: $\|\mathbf{u}\|^2 = 6$.
- The sidelobes measure how much $\mathbf{u}$ resembles its own shifts. Here, quite a lot: at offset 2, $\mathbf{u}$ still overlaps itself by 4.
- A vector that looks like noise to every shifted copy of itself has small sidelobes. A vector with a repeating pattern has large ones.

Keep this picture in mind: when $\mathbf{u}$ is used as a role, the green bar will return the filler, and the red bars will add noise.

Four Facts You Can Check on $(6, 3, 4, 3)$

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

1. **The spike is the largest entry.** A rotated copy has the same length as the original, and a dot product of two vectors of equal length is largest when they are the same vector. So no sidelobe exceeds $\|\mathbf{x}\|^2$: here $3, 4, 3 \leq 6$.
2. **It is symmetric.** Offset j and offset $n - j$ compare the same pairs of entries, once in each direction: offsets 1 and 3 are both 3.
3. **The entries add to the square of the sum.** Every pair of entries $x_k x_m$ is counted once, at offset $m - k$: $6 + 3 + 4 + 3 = 16 = (1 + 2 + 0 + 1)^2$.
4. **A shift does not change it.** Shifting $\mathbf{x}$ moves both copies together, so every overlap stays the same. Our $\mathbf{u}$ is $\mathbf{v}$ shifted one place right, and $\mathbf{v} \# \mathbf{v} = (6, 3, 4, 3)$ too.

Fact 4 says the autocorrelation forgets *where* a pattern sits and keeps *how it repeats*. That is what makes it useful for finding periods.

Four Vectors, Worked by Hand: Constant and Alternating

Same recipe each time: rotate $\mathbf{x}$ left by j, dot with $\mathbf{x}$.

$\mathbf{x} = (1, 1, 1, 1)$

j	rotated left by j	dot with $\mathbf{x}$	
0	$(1, 1, 1, 1)$	$1+1+1+1$	4
1	$(1, 1, 1, 1)$	$1+1+1+1$	4
2	$(1, 1, 1, 1)$	$1+1+1+1$	4
3	$(1, 1, 1, 1)$	$1+1+1+1$	4

$\mathbf{x} \# \mathbf{x} = (4, 4, 4, 4)$

$\mathbf{x} = (1, -1, 1, -1)$

j	rotated left by j	dot with $\mathbf{x}$	
0	$(1, -1, 1, -1)$	$1+1+1+1$	4
1	$(-1, 1, -1, 1)$	$-1-1-1-1$	-4
2	$(1, -1, 1, -1)$	$1+1+1+1$	4
3	$(-1, 1, -1, 1)$	$-1-1-1-1$	-4

$\mathbf{x} \# \mathbf{x} = (4, -4, 4, -4)$

Constant: every rotation is the same vector, so every offset matches perfectly: sidelobes as big as the spike. **Alternating:** an odd rotation flips every sign, so offsets 1 and 3 match perfectly *backward*, -4: sidelobes as big as the spike, negative.

Four Vectors, Worked by Hand: Repeating and Irregular

Same recipe again.

$\mathbf{x} = (1, 0, 1, 0)$

j	rotated left by j	dot with $\mathbf{x}$	
0	$(1, 0, 1, 0)$	$1+0+1+0$	2
1	$(0, 1, 0, 1)$	$0+0+0+0$	0
2	$(1, 0, 1, 0)$	$1+0+1+0$	2
3	$(0, 1, 0, 1)$	$0+0+0+0$	0

$\mathbf{x} \# \mathbf{x} = (2, 0, 2, 0)$

$\mathbf{x} = (1, 2, 0, 1)$

j	rotated left by j	dot with $\mathbf{x}$	
0	$(1, 2, 0, 1)$	$1+4+0+1$	6
1	$(2, 0, 1, 1)$	$2+0+0+1$	3
2	$(0, 1, 1, 2)$	$0+2+0+2$	4
3	$(1, 1, 2, 0)$	$1+2+0+0$	3

$\mathbf{x} \# \mathbf{x} = (6, 3, 4, 3)$

Repeating: the pattern repeats every 2, so the rotation by 2 is the vector itself: one sidelobe as big as the spike; rotating by 1 or 3 puts every 1 on a 0. **Irregular:** our $\mathbf{u}$ repeats nothing exactly, but each rotation still overlaps it somewhat: large sidelobes, 3, 4, 3.

In-Class Exercise: The Fifth Vector

vector	autocorrelation	sidelobes	why
$(1,1,1,1)$	$(4,4,4,4)$	as big as the spike	every shift is the same vector
$(1,-1,1,-1)$	$(4,-4,4,-4)$	as big as the spike	a shift by 1 flips the sign
$(1,0,1,0)$	$(2,0,2,0)$	one as big as the spike	the pattern repeats every 2
$(1,2,0,1)$	$(6,3,4,3)$	large	some accidental overlap
$\frac{1}{2}(1,1,1,-1)$	**?**	**?**	**?**

1. Compute the autocorrelation of $\mathbf{r} = \frac{1}{2}(1,1,1,-1)$ by hand. (Work with $(1,1,1,-1)$, then multiply by $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.)
2. Fill in the last row: how big are its sidelobes, and why?
3. What does this predict for decoding with $\mathbf{r}$ as a role?

Five minutes.

The Fifth Vector: Answers

$$\mathbf{x} = (1, 1, 1, -1)$$

j	**rotated left by** j	**dot with** $\mathbf{x}$	
0	$(1, 1, 1, -1)$	$1 + 1 + 1 + 1$	4
1	$(1, 1, -1, 1)$	$1 + 1 - 1 - 1$	0
2	$(1, -1, 1, 1)$	$1 - 1 + 1 - 1$	0
3	$(-1, 1, 1, 1)$	$-1 + 1 + 1 - 1$	0

$$\mathbf{x} \# \mathbf{x} = (4, 0, 0, 0)$$

1. Times $\frac{1}{4}$: $\mathbf{r} \# \mathbf{r} = (1, 0, 0, 0) = \boldsymbol{\delta}$. **2.** No sidelobes: at every nonzero offset the $+1$s and the -1 cancel exactly, so $\mathbf{r}$ resembles none of its shifts. **3.** Decoding with $\mathbf{r}$ returns the filler exactly: *No Sidelobes, No Error*, below.

Repetition makes sidelobes; a vector unlike all its shifts has none. An autocorrelation of the form (spike, $0, \ldots, 0$) is a multiple of $\boldsymbol{\delta} = (1, 0, \ldots, 0)$, the identity for convolution. The unit impulse $\boldsymbol{\delta}_1 = (0, 1, 0, 0)$ also has autocorrelation $\boldsymbol{\delta}$: shifting it by any nonzero amount moves its single 1 to an empty slot.

Autocorrelation Finds Repetition

Mark *the* in *the dog chased the cat*, one slot per word: it is in slots 0 and 3.

$$\mathbf{x} = (\underbrace{1}_{\textit{the}}, \underbrace{0}_{\textit{dog}}, \underbrace{0}_{\textit{chased}}, \underbrace{1}_{\textit{the}}, \underbrace{0}_{\textit{cat}})$$

A sentence has a start and an end, so here we do not wrap around: offset j pairs slot k only with slot $k + j$ inside the sentence, giving $(2,\ 0,\ 0,\ 1,\ 0)$. An offset is a distance, not a position: the spike, 2, counts the occurrences, and the 1 at offset 3 says one *the* is followed by another three words later.

HRR's own vectors wrap around, and wrapping would change this answer. The next slide shows how, and why.

Where this is used

Speech. A voiced sound repeats with the period of the vocal folds. Pitch trackers, Praat's among them (Boersma 1993), find that period as the offset of the highest sidelobe of the signal's autocorrelation. **Text.** Over a word's positions, it measures how regularly the word recurs.

In signal processing, sidelobes are the information. In HRR they are the noise.

To Wrap or Not to Wrap

Only the two slots holding *the*, 0 and 3, can contribute. At each offset j, ask what each of them finds j places ahead:

j	**from slot 0**	**from slot 3**	**if we wrap, slot 3 finds**	**no wrap**	**wrap**
0	slot 0: *the* ✓	slot 3: *the* ✓		2	2
1	slot 1: *dog*	slot 4: *cat*		0	0
2	slot 2: *chased*	slot 5: past the end	slot 0: *the* ✓	0	1
3	slot 3: *the* ✓	slot 6: past the end	slot 1: *dog*	1	1
4	slot 4: *cat*	slot 7: past the end	slot 2: *chased*	0	0

Without wrapping, $(2, 0, 0, 1, 0)$. With wrapping, $(2, 0, 1, 1, 0)$: the extra 1 at offset 2 comes from running off the end of the sentence and back to its start, where the first *the* is. It is the offset-3 pair read the other way round the circle, $2 = 5 - 3$.

Same vector, two answers. Which one is right depends on whether slot 4 is followed by nothing, or by slot 0.

Why HRR Wraps, and Text Does Not

HRR: a circle

A binding must come back as n numbers, so a shift has to put its last coordinate somewhere, and the only place is the front: $\rho(1, 2, 0, 1) = (1, 1, 2, 0)$. Nothing is lost and nothing added. The coordinates of an HRR vector have no first or last in any meaningful sense, and Session 11c's fold wraps for the same reason.

Text: a line

A sentence has a start and an end. Past *cat* there is nothing, and nothing connects *cat* back to *the*. Wrapping would invent a neighbor relation the sentence does not have: the offset-2 match on the last slide treats the first *the* as if it came two words *after* the second.

On a circle, offsets come in mirror pairs. Going j steps forward around n slots lands where $n - j$ steps backward does, so a wrapped autocorrelation always has offset j equal to offset $n - j$: here offsets 2 and 3, with $5 - 3 = 2$. That is the symmetry fact from earlier. For text, it means each match is reported twice, once from each side.

The practical rule: use the wrapped version for HRR vectors, where the circle is real; use the unwrapped one for sentences and speech, or pad the sequence with enough empty slots that wrapping cannot reach a real word.

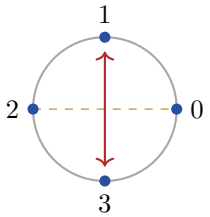
The Involution: Keep the First Entry, Reverse the Rest

index	0	1	2	3
u	1	2	0	1
v	2	0	1	1

Correlation has a partner operation. The **involution** of $\mathbf{x}$ sends slot k to slot $-k$, counted mod n:

$$x^*_k \;=\; x_{(-k) \bmod n}, \qquad \mathbf{x}^* \;=\; (x_0,\; x_{n-1},\; x_{n-2},\; \ldots,\; x_1)$$

slot k	0	1	2	3
reads slot $-k \bmod 4$	0	3	2	1
$\mathbf{u} = (1,2,0,1)$	$\to$	$\mathbf{u}^* = (1,1,0,2)$		
$\mathbf{v} = (2,0,1,1)$	$\to$	$\mathbf{v}^* = ?$ (exercise)		



mirror in the line through slot 0

Think of the slots on a circle and reflect it in the line through slot 0: slot 0 stays, 1 and 3 swap, and here 2 stays too, since $-2 \equiv 2 \pmod 4$. Doing it twice puts everything back, $(\mathbf{x}^*)^* = \mathbf{x}$; a map that is its own inverse is an involution.

Derivation: Correlation Is Convolution With the Involution

Convolve $\mathbf{x}^*$ with $\mathbf{t}$, and rewrite one step at a time (indices mod n):

$$
\begin{aligned}
(\mathbf{x}^* \circledast \mathbf{t})_j &= \sum_k x_k^*\, t_{j-k} && \text{convolution: indices add to } j \\
&= \sum_k x_{-k}\, t_{j-k} && \text{definition of the involution} \\
&= \sum_m x_m\, t_{j+m} && \text{rename } m = -k, \text{ so } j-k = j+m \\
&= (\mathbf{x} \# \mathbf{t})_j && \text{correlation: indices differ by } j
\end{aligned}
$$

The renaming is allowed because as k runs over every slot, so does $m = -k$. Flipping the sign of one index turns "add to j" into "differ by j".

Check one entry

$\mathbf{t} = \mathbf{u} \circledast \mathbf{v} = (4, 5, 2, 5)$, $\mathbf{u}^* = (1, 1, 0, 2)$. Entry 1 of $\mathbf{u}^* \circledast \mathbf{t}$:
$1{\cdot}t_1 + 1{\cdot}t_0 + 0{\cdot}t_3 + 2{\cdot}t_2 = 5 + 4 + 0 + 4 = 13$, the same as $(\mathbf{u} \# \mathbf{t})_1 = 13$.

With $\mathbf{t} = \mathbf{x}$ it gives the autocorrelation: $\mathbf{x} \# \mathbf{x} = \mathbf{x}^* \circledast \mathbf{x}$.

In-Class Exercise: An Involution in the Service of Correlation

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

1. Compute $\mathbf{v}^*$ for $\mathbf{v} = (2, 0, 1, 1)$. Then compute $(\mathbf{v}^*)^*$ and check that you are back at $\mathbf{v}$.
2. Compute $\mathbf{v}^* \circledast \mathbf{v}$ with the convolution formula, $(\mathbf{v}^* \circledast \mathbf{v})_j = \sum_k v_k^* \, v_{j-k}$. Compare it with $\mathbf{v} \# \mathbf{v}$, which you get by rotating $\mathbf{v}$ left and dotting.
3. For a vector of length 5, $\mathbf{x} = (a, b, c, d, e)$, write $\mathbf{x}^*$. Which slots stay put? Which stay put for any even n?
4. $\star$ Plate unbinds with $\mathbf{c}^*$ rather than with $\mathbf{c}$. Using the derivation, say in one sentence what convolving with $\mathbf{c}^*$ computes.

Seven minutes.

The Involution Exercise: Answers

1. $\mathbf{v}^* = (v_0, v_3, v_2, v_1) = (2, 1, 1, 0)$. Again: $(2, 0, 1, 1) = \mathbf{v}$.

2. Products whose indices add to j, with $\mathbf{v}^* = (2, 1, 1, 0)$:

j	$v_0^* v_j + v_1^* v_{j-1} + v_2^* v_{j-2} + v_3^* v_{j-3}$	
0	$2{\cdot}2 + 1{\cdot}1 + 1{\cdot}1 + 0{\cdot}0$	6
1	$2{\cdot}0 + 1{\cdot}2 + 1{\cdot}1 + 0{\cdot}1$	3
2	$2{\cdot}1 + 1{\cdot}0 + 1{\cdot}2 + 0{\cdot}1$	4
3	$2{\cdot}1 + 1{\cdot}1 + 1{\cdot}0 + 0{\cdot}2$	3

$(6, 3, 4, 3) = \mathbf{v} \,\#\, \mathbf{v}$: the same numbers, by convolution with the involution.

3. $\mathbf{x}^* = (a, e, d, c, b)$. Only slot 0 stays put when n is odd; for even n, slots 0 and $n/2$ do.

4. ⋆ Convolving with $\mathbf{c}^*$ correlates with $\mathbf{c}$: it slides $\mathbf{c}$ along the trace and scores the match at every offset, so the filler bound to $\mathbf{c}$ comes back, plus noise from $\mathbf{c}$'s sidelobes.

Why Decoding Depends on the Autocorrelation

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

By the derivation, correlating with $\mathbf{u}$ is convolving with $\mathbf{u}^*$. Convolution is commutative and associative, so the brackets can move:

$$\mathbf{u} \# (\mathbf{u} \circledast \mathbf{v}) \;=\; \mathbf{u}^* \circledast (\mathbf{u} \circledast \mathbf{v}) \;=\; (\mathbf{u}^* \circledast \mathbf{u}) \circledast \mathbf{v} \;=\; (\mathbf{u} \# \mathbf{u}) \circledast \mathbf{v}$$

Decoding convolves the filler with the role's autocorrelation.

Check it on our vectors: $(6,3,4,3) \circledast (2,0,1,1)$, by the convolution formula:

j	**products whose indices add to** j	
0	$6{\cdot}2 + 3{\cdot}1 + 4{\cdot}1 + 3{\cdot}0$	19
1	$6{\cdot}0 + 3{\cdot}2 + 4{\cdot}1 + 3{\cdot}1$	13
2	$6{\cdot}1 + 3{\cdot}0 + 4{\cdot}2 + 3{\cdot}1$	17
3	$6{\cdot}1 + 3{\cdot}1 + 4{\cdot}0 + 3{\cdot}2$	15

The same $(19, 13, 17, 15)$ that decoding gave.

The Spike Returns the Filler; the Sidelobes Add Noise

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Split the autocorrelation into its spike and its sidelobes, and convolve each part with $\mathbf{v}$:

part of $\mathbf{u} \# \mathbf{u}$	**convolved with $\mathbf{v}$**	**what it is**
spike $(6, 0, 0, 0)$	$(12, 0, 6, 6) = 6\,\mathbf{v}$	the filler, scaled by $\|\mathbf{u}\|^2$
sidelobes $(0, 3, 4, 3)$	$(7, 13, 11, 9)$	noise
all of $(6, 3, 4, 3)$	$(19, 13, 17, 15)$	what decoding returns

A spike of 6 at offset 0 is $6\,\boldsymbol{\delta}$, and convolving with $\boldsymbol{\delta}$ changes nothing, so the spike contributes exactly $6\,\mathbf{v}$. The sidelobes contribute shifted copies of $\mathbf{v}$, weighted 3, 4, 3, and those copies are the error.

This is why correlation is only approximate: it returns the filler times the spike, plus the filler smeared by the sidelobes. Decoding is exact exactly when every sidelobe is zero.

No Sidelobes, No Error

Take $\mathbf{r} = \frac{1}{2}(1, 1, 1, -1)$. Its autocorrelation is $(1, 0, 0, 0) = \boldsymbol{\delta}$: a spike of 1 and no sidelobes. So decoding returns the filler exactly:

$$\mathbf{r} \# (\mathbf{r} \circledast \mathbf{v}) \;=\; (\mathbf{r} \# \mathbf{r}) \circledast \mathbf{v} \;=\; \boldsymbol{\delta} \circledast \mathbf{v} \;=\; \mathbf{v}$$

By hand: $\mathbf{r} \circledast \mathbf{v} = (2, 1, 1, 0)$, and correlating that with $\mathbf{r}$ gives $(2, 0, 1, 1) = \mathbf{v}$.

Vectors whose autocorrelation is $\boldsymbol{\delta}$ are the **unitary** vectors: for them, correlation is the exact inverse of binding. The unit impulses are unitary too, which is why a cyclic shift can always be undone exactly.

The catch: unitary vectors are a special, narrow set. Plate's random atoms are not unitary, so they always have sidelobes. The question is how much those sidelobes hurt.

Random Atoms at Large n: Small Sidelobes, but Many

Draw a random atom at $n = 1{,}024$, with variance $1/n$, and compute its autocorrelation:

spike	1.00 ± 0.04
each sidelobe	about ± 0.03, that is $\pm 1/\sqrt{n}$; the largest of them about 0.10
number of sidelobes	$1{,}023$
sum of the squared sidelobes	1.00 ± 0.13

Mean and standard deviation over 200 atoms.

Each sidelobe is tiny, but there are $n - 1$ of them, each with a squared size near $1/n$, so together they carry about as much as the spike. Decoding returns the filler plus one noise piece about as long as the filler, and two equal lengths at right angles give

$$\cos \approx \frac{1}{\sqrt{1+1}} = 0.71 \qquad \text{measured: } 0.71 \pm 0.02$$

Spread over $1{,}024$ directions, that noise looks like no particular atom, so cleanup still finds the filler.

The Fourier View, in One Line

index	0	1	2	3
u	1	2	0	1
v	2	0	1	1

The Fourier transform of an autocorrelation is the squared size of each Fourier coefficient, the **power spectrum**:

$$\mathcal{F}(\mathbf{x} \# \mathbf{x}) \;=\; |\mathcal{F}\mathbf{x}|^2, \qquad \mathcal{F}\mathbf{u} = (4,\ 1{-}i,\ -2,\ 1{+}i) \implies |\mathcal{F}\mathbf{u}|^2 = (16,\ 2,\ 4,\ 2) = \mathcal{F}(6,3,4,3)$$

The autocorrelation and the power spectrum carry the same information, in two coordinate systems. The spike-only autocorrelation $\boldsymbol{\delta}$ transforms to $(1,1,1,1)$:

autocorrelation	**power spectrum**	**decoding**
$\boldsymbol{\delta}$: spike only	flat: every $\lvert\mathcal{F}_k\mathbf{x}\rvert^2 = 1$	exact (unitary)
spike plus sidelobes	uneven, like $(16,2,4,2)$	filler plus noise

In signal processing this pairing is the Wiener–Khinchin theorem. Part V of today's HRR session derives the Fourier transform from scratch; there, the flat spectrum is the test for a unitary role.

Summary

1. The **autocorrelation** of $\mathbf{x}$ lists its overlaps with itself rotated left by j: $(\mathbf{x} \# \mathbf{x})_j = \sum_k x_k x_{k+j}$.
2. Offset 0 is the spike, $\|\mathbf{x}\|^2$, the largest entry. The other offsets are sidelobes: accidental overlaps with shifted copies. The autocorrelation is symmetric, sums to $(\sum_k x_k)^2$, and ignores shifts.
3. Repetition makes large sidelobes; that is how autocorrelation finds periods, in speech and in text.
4. **Decoding convolves the filler with the role's autocorrelation**: $\mathbf{u} \# (\mathbf{u} \circledast \mathbf{v}) = (\mathbf{u} \# \mathbf{u}) \circledast \mathbf{v}$. The spike returns the filler; the sidelobes add noise.
5. The **involution** keeps x_0 and reverses the rest; correlating with $\mathbf{x}$ is convolving with $\mathbf{x}^*$, so $\mathbf{x} \# \mathbf{x} = \mathbf{x}^* \circledast \mathbf{x}$.
6. Roles with no sidelobes, autocorrelation $\boldsymbol{\delta}$, are **unitary**, and decode exactly.
7. A random atom at large n has $n-1$ sidelobes of size about $1/\sqrt{n}$, carrying about as much as the spike: one pair decodes at about 0.71.

Back to the HRR session, at *Where the Error Comes From*.

Exercises

1. Compute $\mathbf{v} \# \mathbf{v}$ for $\mathbf{v} = (2, 0, 1, 1)$ by hand, and say why it equals $\mathbf{u} \# \mathbf{u}$.
2. Compute the autocorrelation of $\mathbf{y} = (1, 1, 0, 0)$. Check facts 1 to 3 on it. Is $\mathbf{y}$ unitary?
3. Use $\mathbf{y}$ as a role: decode $\mathbf{v}$ from $\mathbf{y} \circledast \mathbf{v}$ by computing $(\mathbf{y} \# \mathbf{y}) \circledast \mathbf{v}$. Split the answer into the part from the spike and the part from the sidelobes.
4. In *a rose is a rose is a rose*, mark the slots holding rose. Compute its autocorrelation without wrapping around, and say which offset shows the repetition.
5. $\star$ Why can no sidelobe be larger than the spike? (Hint: a rotated copy has the same length.)

Exercises: Answers

1. $(6, 3, 4, 3)$: offset 0 is $4+0+1+1$, offset 1 is $0+0+1+2$, offset 2 is $2+0+2+0$, offset 3 is $2+0+0+1$. Equal to $\mathbf{u} \# \mathbf{u}$ because $\mathbf{u}$ is $\mathbf{v}$ shifted, and a shift does not change an autocorrelation.

2. $\mathbf{y} \# \mathbf{y} = (2, 1, 0, 1)$. The spike 2 is the largest entry; offsets 1 and 3 agree; and $2+1+0+1 = 4 = (1+1)^2$. It has sidelobes, so it is not unitary.

3. $(2, 1, 0, 1) \circledast \mathbf{v} = (5, 3, 3, 5)$, at cosine 0.89 with $\mathbf{v}$. The spike gives $2\,\mathbf{v} = (4, 0, 2, 2)$; the sidelobes $(0, 1, 0, 1)$ give $(1, 3, 1, 3)$.

4. Eight words, slots 0 to 7, rose in slots 1, 4, 7: $\mathbf{x} = (0, 1, 0, 0, 1, 0, 0, 1)$. Without wrapping, $\mathbf{x} \# \mathbf{x} = (3, 0, 0, 2, 0, 0, 1, 0)$: offset 3 has 2, the pairs of slots $(1, 4)$ and $(4, 7)$, so rose recurs every three words; offset 6 has 1, slots $(1, 7)$, two steps of three.

5. $\star$ A sidelobe is the dot product of $\mathbf{x}$ with a rotated copy of the same length. The dot product of two vectors is at most the product of their lengths, $\|\mathbf{x}\|\,\|\mathbf{x}\| = \|\mathbf{x}\|^2$, which is the spike.

Reading

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